

Exercices on session *Tidyverse 2/2*

For all the following exercises, use functions from the tidyverse, and pipes, whenever possible.

```
library(tidyverse)
```

BCI tree count

You are going to work on data from trees in Barro Colorado Island. These data are available in the package *vegan*, and are called *BCI*. The dataset is a floristic table. It has a row per plot and a column per species. The values are the number of trees of a given species in a given plot.

- Load the data (don't forget to look at the metadata in the help of the package).

```
library(vegan)  
data(BCI)
```

- Get an overview of the data.

```
str(BCI)  
summary(BCI)
```

- Transform the data to a tibble.

```
BCI <- as_tibble(BCI)  
# str(BCI)
```

Transform the dataset

- Add a column with the plot ID (from BCI_1 to BCI_...) (and check what you have done).

```
BCI <- BCI %>%  
  mutate(plot_ID = paste("BCI",  
                          1:nrow(BCI),  
                          sep = "_")  
)
```

```
dim(BCI)
```

```
[1] 50 226
```

```
names(BCI)
```

[1] "Abarema.macradenia"	"Vachellia.melanoceras"
[3] "Acalypha.diversifolia"	"Acalypha.macrostachya"
[5] "Adelia.triloba"	"Aegiphila.panamensis"
[7] "Alchornea.costaricensis"	"Alchornea.latifolia"
[9] "Alibertia.edulis"	"Allophylus.psilospermus"
[11] "Alseis.blackiana"	"Amaioua.corymbosa"
[13] "Anacardium.excelsum"	"Andira.inermis"
[15] "Annona.spraguei"	"Apeiba.glabra"
[17] "Apeiba.tibourbou"	"Aspidosperma.desmanthum"
[19] "Astrocaryum.standleyanum"	"Astronium.graveolens"
[21] "Attalea.butyracea"	"Banara.guianensis"
[23] "Beilschmiedia.pendula"	"Brosimum.alicastrum"
[25] "Brosimum.guianense"	"Calophyllum.longifolium"
[27] "Casearia.aculeata"	"Casearia.arborea"
[29] "Casearia.commersoniana"	"Casearia.guianensis"
[31] "Casearia.sylvestris"	"Cassipourea.guianensis"
[33] "Cavanillesia.platanifolia"	"Cecropia.insignis"
[35] "Cecropia.obtusifolia"	"Cedrela.odorata"
[37] "Ceiba.pentandra"	"Celtis.schippii"
[39] "Cespedesia.spathulata"	"Chamguava.schippii"
[41] "Chimarrhis.parviflora"	"Maclura.tinctoria"
[43] "Chrysochlamys.eclipses"	"Chrysophyllum.argenteum"
[45] "Chrysophyllum.cainito"	"Coccoloba.coronata"
[47] "Coccoloba.manzinellensis"	"Colubrina.glandulosa"

[49] "Cordia.alliodora"	"Cordia.bicolor"
[51] "Cordia.lasiocalyx"	"Coussarea.curvigemma"
[53] "Croton.billbergianus"	"Cupania.cinerea"
[55] "Cupania.latifolia"	"Cupania.rufescens"
[57] "Cupania.seemannii"	"Dendropanax.arboreus"
[59] "Desmopsis.panamensis"	"Diospyros.artanthifolia"
[61] "Dipteryx.oleifera"	"Drypetes.standleyi"
[63] "Elaeis.oleifera"	"Enterolobium.schomburgkii"
[65] "Erythrina.costaricensis"	"Erythroxylum.macrophyllum"
[67] "Eugenia.florida"	"Eugenia.galalonensis"
[69] "Eugenia.nesiotica"	"Eugenia.oerstediana"
[71] "Faramea occidentalis"	"Ficus.colubrinae"
[73] "Ficus.costaricana"	"Ficus.insipida"
[75] "Ficus.maxima"	"Ficus.obtusifolia"
[77] "Ficus.popenoei"	"Ficus.tonduzii"
[79] "Ficus.trigonata"	"Ficus.yoponensis"
[81] "Garcinia.intermedia"	"Garcinia.madruno"
[83] "Genipa.americana"	"Guapira.myrtiflora"
[85] "Guarea.fuzzy"	"Guarea.grandifolia"
[87] "Guarea.guidonia"	"Guatteria.dumetorum"
[89] "Guazuma.ulmifolia"	"Guettarda.foliacea"
[91] "Gustavia.superba"	"Hampea.appendiculata"
[93] "Hasseltia.floribunda"	"Heisteria.acuminata"
[95] "Heisteria.concinna"	"Hirtella.americana"
[97] "Hirtella.triandra"	"Hura.crepitans"
[99] "Hieronyma.alchorneoides"	"Inga.acuminata"
[101] "Inga.cocleensis"	"Inga.goldmanii"
[103] "Inga.laurina"	"Inga.semialata"
[105] "Inga.nobilis"	"Inga.oerstediana"
[107] "Inga.pezizifera"	"Inga.punctata"
[109] "Inga.ruiziana"	"Inga.sapindoides"
[111] "Inga.spectabilis"	"Inga.umbellifera"
[113] "Jacaranda.copaia"	"Lacistema.aggregatum"
[115] "Lacmellea.panamensis"	"Laetia.procera"
[117] "Laetia.thamnia"	"Lafoensia.punicifolia"
[119] "Licania.hypoleuca"	"Licania.platypus"
[121] "Lindackeria.laurina"	"Lonchocarpus.heptaphyllus"
[123] "Luehea.seemannii"	"Macrocnemum.roseum"
[125] "Maquira.guianensis.costaricana"	"Margaritaria.nobilis"
[127] "Marila.laxiflora"	"Maytenus.schippii"
[129] "Miconia.affinis"	"Miconia.argentea"
[131] "Miconia.elata"	"Miconia.hondurensis"
[133] "Mosannonna.garwoodii"	"Myrcia.gatunensis"

[135] "Myrospermum.frutescens"	"Nectandra.cissiflora"
[137] "Nectandra.lineata"	"Nectandra.purpurea"
[139] "Ochroma.pyramidale"	"Ocotea.cernua"
[141] "Ocotea.oblonga"	"Ocotea.puberula"
[143] "Ocotea.whitei"	"Oenocarpus.mapora"
[145] "Ormosia.amazonica"	"Ormosia.coccinea"
[147] "Ormosia.macrocalyx"	"Pachira.quinata"
[149] "Pachira.sessilis"	"Perebea.xanthochyma"
[151] "Cinnamomum.triplinerve"	"Picramnia.latifolia"
[153] "Piper.reticulatum"	"Platymiscium.pinnatum"
[155] "Platypodium.elegans"	"Posoqueria.latifolia"
[157] "Poulsenia.armata"	"Pourouma.bicolor"
[159] "Pouteria.fossicola"	"Pouteria.reticulata"
[161] "Pouteria.stipitata"	"Prioria.copaifera"
[163] "Protium.costaricense"	"Protium.panamense"
[165] "Protium.tenuifolium"	"Pseudobombax.septenatum"
[167] "Psidium.friedrichsthalianum"	"Psychotria.grandis"
[169] "Pterocarpus.rohrii"	"Quararibea.asterolepis"
[171] "Quassia.amara"	"Randia.armata"
[173] "Sapium.broadleaf"	"Sapium.glandulosum"
[175] "Schizolobium.parahyba"	"Senna.dariensis"
[177] "Simarouba.amara"	"Siparuna.guianensis"
[179] "Siparuna.pauciflora"	"Sloanea.terniflora"
[181] "Socratea.exorrhiza"	"Solanum.hayesii"
[183] "Sorocea.affinis"	"Spachea.membranacea"
[185] "Spondias.mombin"	"Spondias.radlkoferi"
[187] "Sterculia.apetala"	"Swartzia.simplex.var.grandiflora"
[189] "Swartzia.simplex.continentalis"	"Symphonia.globulifera"
[191] "Handroanthus.guayacan"	"Tabebuia.rosea"
[193] "Tabernaemontana.arborea"	"Tachigali.versicolor"
[195] "Talisia.nervosa"	"Talisia.princeps"
[197] "Terminalia.amazonia"	"Terminalia.oblonga"
[199] "Tetragastris.panamensis"	"Tetrathylacium.johansenii"
[201] "Theobroma.cacao"	"Thevetia.ahouai"
[203] "Tocoyena.pittieri"	"Trattinnickia.aspera"
[205] "Trema.micrantha"	"Trichanthera.gigantea"
[207] "Trichilia.pallida"	"Trichilia.tuberculata"
[209] "Trichospermum.galeottii"	"Triplaris.cumingiana"
[211] "Trophis.caucana"	"Trophis.racemosa"
[213] "Turpinia occidentalis"	"Unonopsis.pittieri"
[215] "Virola.multiflora"	"Virola.sebifera"
[217] "Virola.surinamensis"	"Vismia.baccifera"
[219] "Vochysia.ferruginea"	"Xylopia.macrantha"

```
[221] "Zanthoxylum.ekmanii"           "Zanthoxylum.juniperinum"
[223] "Zanthoxylum.panamense"         "Zanthoxylum.setulosum"
[225] "Zuelania.guidonia"              "plot_ID"
```

- Create a tibble with the data in a long format, to have one row per species and plot as follow:

```
BCI_L <- BCI %>%
  pivot_longer(!plot_ID,
               names_to = "gen_sp",
               values_to = "abund")
BCI_L
```

```
# A tibble: 11,250 x 3
  plot_ID gen_sp      abund
  <chr>   <chr>      <int>
1 BCI_1  Abarema.macradenia      0
2 BCI_1  Vachellia.melanoceras    0
3 BCI_1  Acalypha.diversifolia    0
4 BCI_1  Acalypha.macrostachya    0
5 BCI_1  Adelia.triloba          0
6 BCI_1  Aegiphila.panamensis     0
7 BCI_1  Alchornea.costaricensis  2
8 BCI_1  Alchornea.latifolia      0
9 BCI_1  Alibertia.edulis         0
10 BCI_1  Allophylus.psilospermus  0
# i 11,240 more rows
```

- Check that the dimensions of this new table are correct (try to do it using a test).

```
nrow(BCI_L) == # number of row in the long format
(ncol(BCI) - 1) * nrow(BCI)
```

```
[1] TRUE
```

```
# number of column (without plot_id) * number of rows in the wide format
```

- We now want to add a column for the genus and species name.
 - Let's do it with the function *separate* (don't replace the initial data, store the result in a new object).

```
BCI_L_2 <- BCI_L %>%
  separate(gen_sp,
    c("genus", "species"), # new column names
    sep = ".",
    remove = FALSE) # keep the gen_sp column
```

Warning: Expected 2 pieces. Additional pieces discarded in 11250 rows [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, ...].

- We get a warning message... It is not very explicit... Let's have a look at the full species name. For that, display the species list. Can we separate the column *gen_sp* based on the separator "." ? Why?

```
sp_list <- BCI_L %>% distinct(gen_sp)
```

```
view(sp_list)
# some gen_sp have more than one . (more than 2 parts),
# so it cannot be separated in the two columns
```

- We want to keep only the characters that are before the second dot (so the genus and the species, removing the variety and subspecies). Let's split the *gen_sp* column by the separator ".". We want to get a matrix with a column by string separated by ".". **NB:** if you say that you want to split by ".", this will be interpreted as "any character", so you need to write "\\." instead.

```
mat_name <- str_split(BCI_L$gen_sp,
  pattern = "\\.", # to split by "." and not by any character
  simplify = TRUE) # to get a matrix and not a list
```

```
view(mat_name)
```

- Let's keep only the two first columns of this matrix and bind these columns to the long dataset.

```
BCI_L <- bind_cols(BCI_L, mat_name[, 1:2])
```

New names:

```
* `` -> `...4`
* `` -> `...5`
```

- Rename the two new columns to have this:

```
BCI_L <- BCI_L %>% rename(
  "genus" = "...4",
  "species" = "...5"
)
BCI_L
```

```
# A tibble: 11,250 x 5
```

	plot_ID <chr>	gen_sp <chr>	abund <int>	genus <chr>	species <chr>
1	BCI_1	Abarema.macradenia	0	Abarema	macradenia
2	BCI_1	Vachellia.melanoceras	0	Vachellia	melanoceras
3	BCI_1	Acalypha.diversifolia	0	Acalypha	diversifolia
4	BCI_1	Acalypha.macrostachya	0	Acalypha	macrostachya
5	BCI_1	Adelia.triloba	0	Adelia	triloba
6	BCI_1	Aegiphila.panamensis	0	Aegiphila	panamensis
7	BCI_1	Alchornea.costaricensis	2	Alchornea	costaricensis
8	BCI_1	Alchornea.latifolia	0	Alchornea	latifolia
9	BCI_1	Alibertia.edulis	0	Alibertia	edulis
10	BCI_1	Allophylus.psilospermus	0	Allophylus	psilospermus

```
# i 11,240 more rows
```

- Get rid of the rows with an abundance of zero.

```
BCI_L <- BCI_L %>%
  filter(abund != 0)
```

Add environmental data

- Let's now add some environmental data from the dataset *BCI.env*. Load it and transform it to a tibble.

```
data(BCI.env)
BCI.env <- as_tibble(BCI.env)
```

- Add a column with the plot ID (from BCI_1 to BCI_...)

```
BCI.env <- BCI.env %>%
  mutate(plot_ID = paste("BCI",
                          1:nrow(BCI.env),
                          sep = "_")
  )
```

- Reorder the levels of *Habitat* from the one with the most plots to the one with the least plots. There is a function for this, it is called *fct_infreq*.

```
# first look at what we have
# levels(BCI.env$Habitat)
# count(BCI.env, Habitat)

BCI.env <- BCI.env %>%
  mutate(Habitat = fct_infreq(BCI.env$Habitat))
```

- Add a column called *EnvHet_cat* taking the following values:
 - *low* when the environmental heterogeneity is lower than 0.08
 - *medium* when the environmental heterogeneity is between 0.08 and 0.49
 - *high* when the environmental heterogeneity is higher than 0.49

```
BCI.env <- BCI.env %>%
  mutate(EnvHet_cat =
    case_when(EnvHet <= 0.08 ~ "low",
              EnvHet > 0.08 & EnvHet <= 0.49 ~ "medium",
              EnvHet > 0.49 ~ "high")
  )
```

- Join the variable *Habitat* and *EnvHet_cat* to the long dataset of abundance (make sure you use the appropriate join function).

```
BCI.env_sub <- BCI.env %>% select(plot_ID, Habitat, EnvHet_cat)

BCI_L <- BCI_L %>%
  left_join(BCI.env_sub,
    by = c("plot_ID" = "plot_ID"))
```